

Zabbix 集群部署指南

该文档基于 Debian13 进行部署测试

架构

节点角色	位置	主机IP	组件	版本	CPU	内存	磁盘	说明
 监控主节点	深圳	10.4.10.169	Zabbix Server	7.4	2核	4GB	20G	负责采集 & 处理监控数据
 监控备节点	香港	10.4.10.170	Zabbix Server	7.4	2核	4GB	20G	HA 冗余
 数据库主库	深圳	10.4.10.171	PostgreSQL + TimescaleDB	16.13 / 2.14.2	4核	8GB	200GB	主写库（高IO）
 数据库备库	香港	10.4.10.172	PostgreSQL + TimescaleDB	16.13 / 2.14.2	4核	8GB	200GB	流复制备库（只读）
 Web 节点	法兰克福	10.4.10.43	Nginx + PHP	-	1核	2GB	20G	Zabbix Web 前端

TimescaleDB 集群部署

修改主机名

```
hostname DataBase

vim /etc/hostname
DataBase
```

安装必要包

```
apt update
apt install -y \
    build-essential cmake git wget curl pkg-config \
    libreadline-dev zlib1g-dev flex bison \
    libssl-dev libxml2-dev libxslt1-dev \
    libkrb5-dev libicu-dev
```

编译安装 PostgreSQL 16.13

```
# 下载源码
wget https://ftp.postgresql.org/pub/source/v16.13/postgresql-16.13.tar.gz
tar -xzf postgresql-16.13.tar.gz
cd postgresql-16.13
# 编译安装
./configure --prefix=/usr/local/pgsql \
            --with-openssl \
            --with-icu \
            --with-libxml \
            --with-libxslt

make -j$(nproc)
make install
# 创建用户 & 数据目录
useradd -m postgres
mkdir -p /data/pgdata
chown -R postgres:postgres /data/pgdata
# 初始化数据库
sudo -u postgres /usr/local/pgsql/bin/initdb -D /data/pgdata
# 配置环境变量
echo 'export PATH=/usr/local/pgsql/bin:$PATH' | sudo tee /etc/profile.d/pgsql.sh
source /etc/profile
# 启动 PostgreSQL
sudo -u postgres /usr/local/pgsql/bin/pg_ctl -D /data/pgdata -l logfile start
# 验证
psql -U postgres -c "SELECT version();"
```

编译安装 TimescaleDB 2.14.2

```
# 下载源码
git clone https://github.com/timescale/timescaledb
cd timescaledb
git checkout 2.14.2
# 指定 PostgreSQL 路径编译
./bootstrap -DREGRESS_CHECKS=OFF \
            -DPG_CONFIG=/usr/local/pgsql/bin/pg_config
```

```
cd build
make -j$(nproc)
sudo make install
```

配置 PostgreSQL 启用 TimescaleDB

```
# 修改配置
vim /data/pgdata/postgresql.conf
##### 添加下面配置 #####
shared_preload_libraries = 'timescaledb'

# 重启数据库
sudo -u postgres /usr/local/pgsql/bin/pg_ctl -D /data/pgdata restart

# 创建扩展
psql -U postgres # 进入数据库
##### 执行 #####
postgres=# CREATE EXTENSION timescaledb;
postgres=# \dx # 验证
postgres=# timescaledb-tune --pg-config=/usr/local/pgsql/bin/pg_config # 性能调优

# 关闭数据库
sudo -u postgres /usr/local/pgsql/bin/pg_ctl -D /data/pgdata stop
```

设置 systemd 服务

```
sudo nano /etc/systemd/system/postgresql.service
##### 添加下面内容 #####
[Unit]
Description=PostgreSQL database server
After=network.target

[Service]
Type=forking
User=postgres
```

```
ExecStart=/usr/local/pgsql/bin/pg_ctl -D /data/pgdata -l /data/pgdata/logfile
start
ExecStop=/usr/local/pgsql/bin/pg_ctl -D /data/pgdata stop
ExecReload=/usr/local/pgsql/bin/pg_ctl -D /data/pgdata reload
PIDFile=/data/pgdata/postmaster.pid
```

```
[Install]
```

```
WantedBy=multi-user.target
```

```
# 开机自启
```

```
sudo systemctl daemon-reexec
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl restart postgresql
```

zabbix数据导入

```
sudo -u postgres /usr/local/pgsql/bin/createuser --pwprompt zabbix # 创建数据库用户
```

```
sudo -u postgres /usr/local/pgsql/bin/createdb -O zabbix zabbix # 创建数据库
```

```
sudo -u postgres /usr/local/pgsql/bin/psql -d zabbix -c "CREATE EXTENSION
timescaledb;" # 启用 TimescaleDB 扩展
```

```
# 在/database/postgresql路径下执行下面代码
```

```
sudo -u postgres /usr/local/pgsql/bin/psql -d zabbix -f schema.sql
```

```
sudo -u postgres /usr/local/pgsql/bin/psql -d zabbix -f images.sql
```

```
sudo -u postgres /usr/local/pgsql/bin/psql -d zabbix -f data.sql
```

```
sudo -u postgres /usr/local/pgsql/bin/psql -d zabbix -f timescaledb/schema.sql
```

```
# 配置权限
```

```
root@DataBase:~# psql -U postgres -d zabbix
```

```
psql (16.13)
```

```
Type "help" for help.
```

```
GRANT SELECT ON ALL TABLES IN SCHEMA public TO zabbix;
```

```
ALTER DEFAULT PRIVILEGES IN SCHEMA public
```

```
GRANT SELECT ON TABLES TO zabbix;
```

```
DO $$
```

```
DECLARE r RECORD;
```

```
BEGIN
```

```

FOR r IN SELECT tablename FROM pg_tables WHERE schemaname = 'public'
LOOP
    EXECUTE 'ALTER TABLE public.' || quote_ident(r.tablename) || ' OWNER TO
zabbix';
END LOOP;
END $$;

-- 所有序列
DO $$
DECLARE r RECORD;
BEGIN
    FOR r IN SELECT sequence_name FROM information_schema.sequences WHERE
sequence_schema='public'
    LOOP
        EXECUTE 'ALTER SEQUENCE public.' || quote_ident(r.sequence_name) || ' OWNER
TO zabbix';
    END LOOP;
END $$;

-- 再补一遍权限
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA public TO zabbix;
GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA public TO zabbix;

```

主库配置

修改配置

```

nano /data/pgdata/postgresql.conf
#### 添加下面内容 #####
listen_addresses = '*'

wal_level = replica
max_wal_senders = 10
max_replication_slots = 10

wal_keep_size = 1GB
hot_standby = on

```

配置访问权限

```
nano /data/pgdata/pg_hba.conf
##### 添加下面内容 #####

# 复制用户
host replication replicator 10.4.10.172/32 md5

# Zabbix访问（按需）
host all all 10.0.0.0/8 md5
```

创建复制用户

```
sudo -u postgres /usr/local/pgsql/bin/psql

CREATE ROLE replicator WITH REPLICATION LOGIN PASSWORD 'Catixs@3202';
```

重启主库

```
systemctl restart postgresql
```

从库配置

停止从库

```
systemctl stop postgresql
```

清空数据目录

```
rm -rf /data/pgdata/*
```

从主库拉取数据

```
pg_basebackup -h 10.4.10.171 -U replicator -D /data/pgdata -Fp -Xs -P -R
```

修改权限

```
chown -R postgres:postgres /data/pgdata
```

启动从库

```
systemctl start postgresql
```

主从验证

```
# 主库执行
SELECT client_addr, state FROM pg_stat_replication;

# 从库执行
SELECT pg_is_in_recovery();
```

Zabbix Server 集群部署

设置主机名

```
hostname ZabbixServer

vim /etc/hostname
##### 添加下面内容 #####
ZabbixServer
```

安装Zabbix 存储库

```
wget https://repo.zabbix.com/zabbix/7.4/release/debian/pool/main/z/zabbix-
release/zabbix-release_latest_7.4+debian13_all.deb
dpkg -i zabbix-release_latest_7.4+debian13_all.deb
apt update
```

安装zabbix 服务器


```
apt install -y \
zabbix-server-pgsql \
zabbix-sql-scripts \
zabbix-agent
```

server 配置

```
#高可用主要配置 HANodeName 和 NodeAddress
#HANodeName: 名称必须唯一, 这是将在 agent 和 proxy 配置中引用 server 的名称
#NodeAddress: 将被 Zabbix 前端用来连接到主 server 节点
root@ZabbixServer:~# grep -v '^#' /etc/zabbix/zabbix_server.conf |grep -v '^$'
LogFile=/var/log/zabbix/zabbix_server.log
LogFileSize=0
PidFile=/run/zabbix/zabbix_server.pid
SocketDir=/run/zabbix
DBHost=10.4.10.171
DBName=zabbix
DBUser=zabbix
DBPassword=Catixs@3202
StartPingers=5
StartVMwareCollectors=2
VMwareCacheSize=64M
SNMPTrapperFile=/var/log/snmptrap/snmptrap.log
CacheSize=128M
HistoryCacheSize=64M
HistoryIndexCacheSize=64M
TrendCacheSize=64M
Timeout=4
FpingLocation=/usr/bin/fping
Fping6Location=/usr/bin/fping6
LogSlowQueries=3000
StatsAllowedIP=127.0.0.1

EnableGlobalScripts=0
HANodeName=master # 不同集群名称需要不一样
NodeAddress=10.4.10.169:10051 # 填写本机地址
```

```
Include=/etc/zabbix/zabbix_server.d/*.conf
```

agent 配置

```
root@ZabbixServer:~# grep -v '^#' /etc/zabbix/zabbix_agentd.conf |grep -v '^$'
PidFile=/run/zabbix/zabbix_agentd.pid
LogFile=/var/log/zabbix/zabbix_agentd.log
LogFileSize=0
Server=127.0.0.1,master,slave
ServerActive=127.0.0.1,master,slave
Hostname=Zabbix server
Include=/etc/zabbix/zabbix_agentd.d/*.conf
```

启动服务

```
systemctl enable zabbix-server
systemctl enable zabbix-agent
systemctl start zabbix-server
systemctl start zabbix-agent
```

Web 节点部署

修改主机名

```
hostname Zabbix-Web
echo 'Zabbix-Web' > /etc/hostname
```

添加中文包

```
vim /etc/locale.gen  
### 取消下面注释 #####  
zh_CN.UTF-8 UTF-8
```

```
locale-gen
```

安装组件

```
wget https://repo.zabbix.com/zabbix/7.4/release/debian/pool/main/z/zabbix-  
release/zabbix-release_latest_7.4+debian13_all.deb  
dpkg -i zabbix-release_latest_7.4+debian13_all.deb  
apt update  
  
apt install -y \  
nginx \  
php-fpm \  
php8.4-pgsql \  
zabbix-frontend-php \  
zabbix-nginx-conf
```

配置 Nginx

```
vim /etc/zabbix/nginx.conf  
##### 修改为如下配置 #####  
listen 80;  
server_name 10.4.10.43;
```

启动服务

```
systemctl restart nginx php8.4-fpm  
systemctl enable nginx php8.4-fpm
```